

45. Selection aur Range

Document me selection (aur `<input>` jaise form fields me selection) ko JS se padh sakte hain, set kar sakte hain, hata sakte hain, tag me wrap kar sakte hain, etc.

Range

Selection ka basic concept: do **boundary points** (start aur end).

js

```
let range = new Range();
range.setStart(node, offset);
range.setEnd(node, offset);
```

`node` text node ya element node ho sakta hai, aur `offset` ka matlab uske hisaab se badalta hai:

- **Text node:** `offset` = text me **position** (characters).

js

```
// <p id="p">Hello</p>
range.setStart(p.firstChild, 2);
range.setEnd(p.firstChild, 4);
console.log(range); // "ll"
```

- **Element node:** `offset` = **child ka number** (poore nodes select karne ke liye).

js

```
// <p id="p">Example: <i>italic</i> and <b>bold</b></p>
range.setStart(p, 0);
range.setEnd(p, 2); // 2 tak, 2 include nahi
// range = "Example: italic"
```

Start aur end alag alag nodes me bhi ho sakte hain (bas end, start ke baad ho).

Range properties

- `startContainer`, `startOffset`
- `endContainer`, `endOffset`
- `collapsed`: start = end (koi content nahi)
- `commonAncestorContainer`: sab nodes ka sabse paas ka common ancestor

Range set karne ke methods

- `setStart(node, offset)`, `setStartBefore(node)`, `setStartAfter(node)`
- `setEnd(node, offset)`, `setEndBefore(node)`, `setEndAfter(node)`
- `selectNode(node)`: poora node select
- `selectNodeContents(node)`: node ke andar ka poora content
- `collapse(toStart)`: range ko collapse karo
- `cloneRange()`: same start/end ka naya range

Range editing methods

- `deleteContents()`: content document se hatao
- `extractContents()`: hatao aur `DocumentFragment` me lo
- `cloneContents()`: clone karke `DocumentFragment` me lo
- `insertNode(node)`: range ki shuruaat me node daalo
- `surroundContents(node)`: node me content ko wrap karo (range me dono opening aur closing tags hone chahiye, `<i>abc`) jaise partial nahi

Selection

`Range` sirf ek object hai, isse screen par kuch select nahi hota. Document ka selection

`Selection` object hai: `window.getSelection()` ya `document.getSelection()`.

- Theory me ek se zyada ranges ho sakti hain (sirf **Firefox** me `Ctrl+click` se), baaki browsers me max 1.

Selection properties

- `anchorNode`, `anchorOffset`: selection **kahan se shuru**
- `focusNode`, `focusOffset`: kahan **khatam**
- `isCollapsed`: kuch select nahi
- `rangeCount`: ranges ki ginti (Firefox ke alawa max 1)
- `getRangeAt(i)`: i-th range

Farak: `Range` ka start hamesha end se pehle hota hai, par selection me nahi. Mouse se left-to-right ya **right-to-left** dono tarah select ho sakta hai, isliye `focus`, `anchor` se pehle bhi ho sakta hai.

Selection events

- `elem.onselctstart`: selection is element par **shuru** ho. Isme default action roko to yaha se selection shuru nahi hoga.
- `document.onselectionchange`: selection badle ya shuru ho (sirf `document` par).

Copy karna

1. Text: `document.getSelection().toString()`

2. DOM (formatting ke saath): `selection.getRangeAt(i).cloneContents()`
(DocumentFragment)

Selection methods

- `addRange(range)`, `removeRange(range)`, `removeAllRanges()`, `empty()`
- `collapse(node, offset)` / `setPosition`, `collapseToStart()`, `collapseToEnd()`
- `extend(node, offset)`: focus ko hilao
- `setBaseAndExtent(anchorNode, anchorOffset, focusNode, focusOffset)`: selection set karo
- `selectAllChildren(node)`
- `deleteFromDocument()`: selected content hatao
- `containsNode(node, allowPartialContainment)`

Poora `<p>` ka content select:

```
js
document.getSelection().setBaseAndExtent(p, 0, p, p.childNodes.length);
// ya
let range = new Range();
range.selectNodeContents(p);
document.getSelection().removeAllRanges(); // pehle purana hatao
document.getSelection().addRange(range);
```

Naya select karne se pehle `removeAllRanges()` karo, warna Firefox ke alawa browsers naya range ignore kar dete hain. (`setBaseAndExtent` jaise methods jo khud replace karte hain, unme zarurat nahi.)

Form controls me selection (`input`, `textarea`)

Alag simple API (kyunki value plain text hai):

- `input.selectionStart`, `selectionEnd` (writable), `selectionDirection`
- Event: `input.onselect`
- Methods:
 - `input.select()`: sab select
 - `setSelectionRange(start, end, [direction])`
 - `setRangeText(replacement, [start], [end], [selectionMode])`: text replace.
`selectionMode`: `"select"`, `"start"`, `"end"`, `"preserve"` (default).

Cursor move karna: `selectionStart = selectionEnd = 10` (dono barabar = cursor us position par).

```
js
area.onfocus = () => {
  setTimeout(() => { area.selectionStart = area.selectionEnd = 10; }); // zero
};
```

Selection ko `(*...*)` me wrap:

```
js
let selected = input.value.slice(input.selectionStart, input.selectionEnd);
input.setRangeText(`*${selected}*`);
```

Cursor par text daalna:

```
js
input.setRangeText("HELLO", input.selectionStart, input.selectionEnd, "end");
input.focus();
```

Unselectable banana (3 tarike)

1. CSS: `(user-select: none)` (selection yaha se shuru nahi hoga, par dusri jagah se extend karke shamil ho sakta hai, copy me aam taur par ignore hota hai)
2. `(onselectstart)` ya `(mousedown)` me default action roko (`(elem.onselectstart = () => false)`)
3. Baad me `(document.getSelection().empty())` (kam use, blink karta hai)

Summary

- Document: `(Selection)` + `(Range)` objects. Form fields: `(selectionStart/End)`, `(setRangeText)` etc.
- Selection lena: `(getSelection())` aur `(getRangeAt(i).cloneContents())`.
- Selection set karna: `(setBaseAndExtent(...))` ya `(removeAllRanges())` + `(addRange(range))`.
- Cursor position = selection ka start/end.